Q1. JEE Main 2026 (21 January Shift 2)

Consider the following spectral lines for atomic hydrogen :
A. First line of Paschen series
B. Second line of Balmer series
C. Third line of Paschen series
D. Fourth line of Bracket series.
The correct arrangement of the above lines in ascending order of energy is :

(1) $D < C < A < B$
(2) $D < A < C < B$
(3) $C < D < B < A$
(4) $A < B < C < D$

Q2. JEE Main 2026 (23 January Shift 1)

Given,
(A) $n = 5,\ m_1 = -1$
(B) $n = 3, l = 2,\ m_1 = -1,\ m_s = +\frac{1}{2}$
The maximum number of electron(s) in an atom that can have the quantum numbers as given in (A) and (B) respectively are :

(1) 8 and 1
(2) 2 and 4
(3) 26 and 1
(4) 4 and 1

Q3. JEE Main 2026 (23 January Shift 1)

Which of the following statements regarding the energy of the stationary state is true in the following one - electron systems ?

(1) $+2.18 \times 10^{-18}$ J for second orbit of He^{+}ion
(2) -2.18×10^{-18} J for third orbit of Li^{2+} ion
(3) -1.09×10^{-18} J for second orbit of H atom.
(4) $+8.72 \times 10^{-18}$ J for first orbit of He^{+}ion

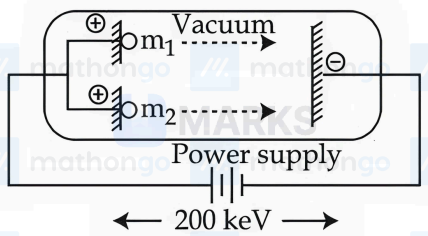
Q4. JEE Main 2026 (28 January Shift 2)

The wavelength of photon ' A ' is 400 nm. The frequency of photon ' B ' is 10^{16} s^{-1}. The wave number of photon ' C' is 10^{4} cm^{-1}. The correct order of energy of these photons is :

(1) $C > B > A$
(2) $B > A > C$
(3) $A > B > C$
(4) $A > C > B$

Q5. JEE Main 2026 (28 January Shift 2)

Two positively charged particles m_1 and m_2 have been accelerated across the same potential difference of 200 keV as shown below.



[Given mass of $m_1 = 1$ amu and $m_2 = 4$ amu]

The deBroglie wavelength of m_1 will be x times of m_2. The value of x is ____ (nearest integer)

Q6. JEE Main 2026 (24 January Shift 2)

The wavelength of spectral line obtained in the spectrum of Li^{2+} ion, when the transition takes place between two levels whose sum is 4 and difference is 2, is

(1) 1.14×10^{-7} cm
(2) 1.14×10^{-6} cm
(3) 2.28×10^{-7} cm
(4) 2.28×10^{-6} cm

Q7. JEE Main 2026 (22 January Shift 2)

The energy of first (lowest) Balmer line of H atom is x J. The energy (in J) of second Balmer line of H atom is :

(1) $1.35x$
(2) x^2
(3) $2x$
(4) $\frac{x}{1.35}$

Q8. JEE Main 2026 (22 January Shift 1)

The energy required by electrons, present in the first Bohr orbit of hydrogen atom to be excited to second Bohr orbit is ____ $Jmol^{-1}$. Given: $R_H = 2.18 \times 10^{-11}$ergs.

(1) 9.835×10^{12}
(2) 1.635×10^{-18}
(3) 1.635×10^{-11}
(4) 9.835×10^{5}

Q9. JEE Main 2026 (23 January Shift 2)

The work functions of two metals (M_A and M_B) are in the 1 : 2 ratio. When these metals are exposed to photons of energy 6 eV, the kinetic energy of liberated electrons of $M_A : M_B$ is in the ratio of 2.642 : 1. The work functions (in eV) of M_A and M_B are respectively.

(1) $1.4, 2.8$
(2) $2.3, 4.6$
(3) $1.5, 3.0$
(4) $3.1, 6.2$

Q10. JEE Main 2026 (23 January Shift 2)

Identify the INCORRECT statements from the following:

A. Notation $^{24}_{12}\mathrm{Mg}$ represents 24 protons and 12 neutrons.

B. Wavelength of a radiation of frequency $4.5 \times 10^{15}\ \mathrm{s}^{-1}$ is 6.7×10^{-8} m.

C. One radiation has wavelength $= \lambda_1(900\ \mathrm{nm})$ and energy $= \mathrm{E}_1$. Other radiation has wavelength $= \lambda_2(300\ \mathrm{nm})$ and energy $= \mathrm{E}_2$. $\mathrm{E}_1 : \mathrm{E}_2 = 3 : 1$.

D. Number of photons of light of wavelength 2000 pm that provides 1 J of energy is 1.006×10^{16}.

Choose the correct answer from the options given below:

(1) B and C Only
(2) A and D Only
(3) A and C Only
(4) A and B Only

Q11. JEE Main 2026 (21 January Shift 1)

Given below are two statements :

Statement I: When an electric discharge is passed through gaseous hydrogen, the hydrogen molecules dissociate and the energetically excited hydrogen atoms produce electromagnetic radiation of discrete frequencies.

Statement II: The frequency of second line of Balmer series obtained from He^+ is equal to that of first line of Lyman series obtained from hydrogen atom.

In the light of the above statements, choose the correct answer from the options given below :

(1) Statement I is false but Statement II is true
(2) Both Statement I and Statement II are true
(3) Statement I is true but Statement II is false
(4) Both Statement I and Statement II are false

Q12. JEE Main 2026 (28 January Shift 1)

The wave numbers of three spectral lines of H atom are considered. Identify the set of spectral lines belonging to Balmer series.

(R = Rydberg constant)

(1) $\frac{7\mathrm{R}}{144}, \frac{3\mathrm{R}}{16}, \frac{16\mathrm{R}}{255}$
(2) $\frac{5\mathrm{R}}{36}, \frac{8\mathrm{R}}{9}, \frac{15\mathrm{R}}{16}$
(3) $\frac{3\mathrm{R}}{4}, \frac{3\mathrm{R}}{16}, \frac{7\mathrm{R}}{144}$
(4) $\frac{5\mathrm{R}}{36}, \frac{3\mathrm{R}}{16}, \frac{21\mathrm{R}}{100}$

Q13. JEE Main 2026 (28 January Shift 1)

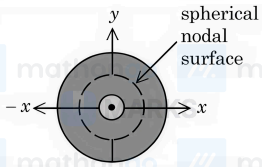


Figure 1. electron probability density for 2s orbital

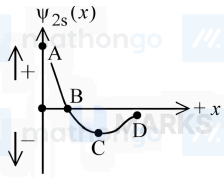


Figure 2. wave function for 2s orbital

Which of the following point in Figure 2 most accurately represents the nodal surface as shown in Figure 1?

(1) D (2) A (3) C (4) B

Q14. JEE Main 2026 (24 January Shift 1)

The hydrogen spectrum consists of several spectral lines in Lyman series ($L_1, L_2, L_3 \ldots$; L_1 has lowest energy among Lyman series). Similarly it consists of several spectral lines in Balmer series ($B_1, B_2, B_3 \ldots$; B_1 has lowest energy among Balmer lines). The energy of L_1 is x times the energy of B_1. The value of x is ____ $\times 10^{-1}$. (Nearest integer)

ANSWERS AND SOLUTIONS

1. (2)	**2.** (1)	**3.** (2)	**4.** (2)	**5.** 2	**6.** (2)	**7.** (1)	**8.** (4)
9. (2)	**10.** (3)	**11.** (2)	**12.** (4)	**13.** (4)	**14.** 54		

1. **(2)** The energy of a spectral line in the hydrogen atom is given by $E = 13.6\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$ eV.

A. First line of Paschen series: $n_1 = 3, n_2 = 4$.
$E_A = 13.6\left(\frac{1}{3^2} - \frac{1}{4^2}\right) = 13.6\left(\frac{1}{9} - \frac{1}{16}\right) = 13.6 \times \frac{7}{144} \approx 0.66$ eV.

B. Second line of Balmer series: $n_1 = 2, n_2 = 4$.
$E_B = 13.6\left(\frac{1}{2^2} - \frac{1}{4^2}\right) = 13.6\left(\frac{1}{4} - \frac{1}{16}\right) = 13.6 \times \frac{3}{16} = 2.55$ eV.

C. Third line of Paschen series: $n_1 = 3, n_2 = 6$.
$E_C = 13.6\left(\frac{1}{3^2} - \frac{1}{6^2}\right) = 13.6\left(\frac{1}{9} - \frac{1}{36}\right) = 13.6 \times \frac{3}{36} = 13.6 \times \frac{1}{12} \approx 1.13$ eV.

D. Fourth line of Brackett series: $n_1 = 4, n_2 = 8$.
$E_D = 13.6\left(\frac{1}{4^2} - \frac{1}{8^2}\right) = 13.6\left(\frac{1}{16} - \frac{1}{64}\right) = 13.6 \times \frac{3}{64} \approx 0.6375$ eV.

Comparing the values: $E_D(0.6375) < E_A(0.66) < E_C(1.13) < E_B(2.55)$.
The ascending order of energy is $D < A < C < B$.

2. **(1)** For case (A) with n=5, $m_l = -1$:
The principal quantum number n=5 allows l from 0 to 4. Since $m_l = -1$, we need l≥1. For each l value (1, 2, 3, 4), the orbital $m_l = -1$ exists, giving 4 orbitals. Each orbital holds 2 electrons (spin up and down), so maximum electrons = 8.
For case (B) with n=3, l=2, $m_l = -1$, $m_s = +\frac{1}{2}$:
These quantum numbers completely specify a single electron in a unique orbital with a specific spin, so maximum electrons = 1.
Therefore, the answer is 8 and 1.

3. **(2)** The energy of an electron in the nth stationary state of a one-electron system is given by the Bohr's formula:
$E_n = -2.18 \times 10^{-18} \times \frac{Z^2}{n^2}$ J
where Z is the atomic number and n is the principal quantum number.
For option (1): Second orbit of He^+ $(Z = 2, n = 2)$
$E = -2.18 \times 10^{-18} \times \frac{2^2}{2^2} = -2.18 \times 10^{-18}$ J. (Incorrect sign in option)
For option (2): Third orbit of Li^{2+} $(Z = 3, n = 3)$

$E = -2.18 \times 10^{-18} \times \frac{3^2}{3^2} = -2.18 \times 10^{-18}$ J. (Correct)

For option (3): Second orbit of H atom ($Z = 1, n = 2$)

$E = -2.18 \times 10^{-18} \times \frac{1^2}{2^2} = -0.545 \times 10^{-18}$ J. (Incorrect value)

For option (4): First orbit of He^+ ($Z = 2, n = 1$)

$E = -2.18 \times 10^{-18} \times \frac{2^2}{1^2} = -8.72 \times 10^{-18}$ J. (Incorrect sign in option)

4. **(2)** Photon A: $\lambda_A = 400$ nm

$E_A = \frac{hc}{\lambda_A} = \frac{6.626\times10^{-34}\times3\times10^8}{400\times10^{-9}} = 4.97 \times 10^{-19}$ J

Photon B: $\nu_B = 10^{16}$ s^{-1}

$E_B = h\nu_B = 6.626 \times 10^{-34} \times 10^{16} = 6.626 \times 10^{-18}$ J

Photon C: $\bar{\nu}_C = 10^4$ cm^{-1} = 10^6 m^{-1}

$\lambda_C = 10^{-6}$ m

$E_C = \frac{hc}{\lambda_C} = \frac{6.626\times10^{-34}\times3\times10^8}{10^{-6}} = 1.988 \times 10^{-19}$ J

Order: $E_B > E_A > E_C$

5. **(2)** For a charged particle accelerated through potential V, the de Broglie wavelength is $\lambda = \frac{h}{\sqrt{2mVe}}$.

The wavelength is inversely proportional to $\sqrt{m}$.

Therefore: $\frac{\lambda_1}{\lambda_2} = \sqrt{\frac{m_2}{m_1}} = \sqrt{\frac{4}{1}} = 2$.

This means $\lambda_1 = 2\lambda_2$, so x = 2 (nearest integer).

6. **(2)** For Li^{2+} (Z=3): From $n_2 + n_1 = 4$ and $n_2 - n_1 = 2$, we get $n_2 = 3$, $n_1 = 1$

Using Bohr wavelength formula: $\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

$\frac{1}{\lambda} = 1.097 \times 10^7 \times 9 \times \left(1 - \frac{1}{9}\right) = 9.873 \times 10^7 \times \frac{8}{9}$

$\frac{1}{\lambda} = 8.776 \times 10^7$ m^{-1}

$\lambda = 1.14 \times 10^{-6}$ cm

7. **(1)** First Balmer line:

transition n=3 to n=2, energy $E_1 = 13.6(\frac{1}{4} - \frac{1}{9}) = 13.6 \times \frac{5}{36} = x$ J.

Second Balmer line:

transition n=4 to n=2, energy $E_2 = 13.6(\frac{1}{4} - \frac{1}{16}) = 13.6 \times \frac{3}{16}$ J.

Ratio: $\frac{E_2}{E_1} = \frac{\frac{3}{16}}{\frac{5}{36}} = \frac{3}{16} \times \frac{36}{5} = \frac{27}{20} = 1.35$.

Thus second Balmer line energy is $1.35x$ J.

8. **(4)** The energy of an electron in the nth Bohr orbit of a hydrogen atom is given by $E_n = -R_H\left(\frac{1}{n^2}\right)$.

The energy required for excitation from $n = 1$ to $n = 2$ for a single atom is

$\Delta E = E_2 - E_1 = R_H\left(\frac{1}{1^2} - \frac{1}{2^2}\right) = R_H\left(1 - \frac{1}{4}\right) = \frac{3}{4}R_H$.

Given $R_H = 2.18 \times 10^{-11}$ ergs.

Converting ergs to Joules: 1 erg $= 10^{-7}$ J.
So, $R_H = 2.18 \times 10^{-11} \times 10^{-7} = 2.18 \times 10^{-18}$ J.
The energy required per atom is $\Delta E = \frac{3}{4} \times 2.18 \times 10^{-18} = 1.635 \times 10^{-18}$ J.
To find the energy required per mole of electrons, multiply by Avogadro's number $N_A \approx 6.022 \times 10^{23}$ mol^{-1}.
Energy per mole $= 1.635 \times 10^{-18} \times 6.022 \times 10^{23}$ J mol^{-1}.
Energy per mole $\approx 9.846 \times 10^5$ J mol^{-1}.
Comparing with the given options, 9.835×10^5 is the closest value.

9. **(2)** The work functions satisfy $W_A : W_B = 1 : 2$.
Let $W_A = W$ and $W_B = 2W$.
Using the photoelectric equation
$KE = h\nu - W$, we have $KE_A = 6 - W$ and $KE_B = 6 - 2W$.
Given the kinetic energy ratio $\frac{KE_A}{KE_B} = 2.642$, we get $\frac{6-W}{6-2W} = 2.642$.
Solving: $6 - W = 2.642(6 - 2W)$ gives $6 - W = 15.852 - 5.284W$
So $4.284W = 9.852$ and $W = 2.3$ eV.
Thus $W_A = 2.3$ eV and $W_B = 4.6$ eV.

10. **(3)** Checking each statement:
(A) states $^{24}_{12}Mg$ has 24 protons, but the mass number is 24 and atomic number is 12, so this is INCORRECT.
(B) Using $\lambda = \frac{c}{f} = \frac{3\times10^8}{4.5\times10^{15}} = 6.67 \times 10^{-8}$ m, which is CORRECT.
(C) For $\lambda_1 = 900$ nm and $\lambda_2 = 300$ nm, since energy is inversely proportional to wavelength, $E_1 : E_2 = 1 : 3$, not $3 : 1$ as stated, so this is INCORRECT.
(D) For photons at $\lambda = 2000$ pm, energy per photon is $E = \frac{hc}{\lambda} = 9.94 \times 10^{-17}$ J, giving $N = 1.006 \times 10^{16}$ photons for 1 J, which is CORRECT.
The incorrect statements are A and C.

11. **(2)** Statement I: Electric discharge through H_2 causes dissociation into excited H atoms which emit discrete frequencies (line spectrum). This is true.
Statement II: Using $\nu = RcZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$
For He^+ (Z=2), second Balmer line ($n_1 = 2, n_2 = 4$): $\nu = 4Rc\left(\frac{1}{4} - \frac{1}{16}\right) = \frac{3Rc}{4}$
For H (Z=1), first Lyman line ($n_1 = 1, n_2 = 2$): $\nu = Rc\left(1 - \frac{1}{4}\right) = \frac{3Rc}{4}$
Both frequencies are equal. Statement II is true.
Both statements are true.

12. **(4)** For the Balmer series, transitions end at n=2.
Using the Rydberg formula $\bar{\nu} = R\left(\frac{1}{2^2} - \frac{1}{n^2}\right) = R\left(\frac{1}{4} - \frac{1}{n^2}\right)$, we calculate wavenumbers for transitions from different upper levels:
From n=3: $\bar{\nu} = R\left(\frac{1}{4} - \frac{1}{9}\right) = \frac{5R}{36}$

From n=4: $\bar{\nu} = R\left(\frac{1}{4} - \frac{1}{16}\right) = \frac{3R}{16}$

From n=5: $\bar{\nu} = R\left(\frac{1}{4} - \frac{1}{25}\right) = \frac{21R}{100}$

Option (4) contains exactly these three wavenumbers corresponding to the Balmer series transitions.

13. (4) The 2s orbital has one radial nodal surface, which is a spherical surface shown in Figure 1.

A nodal surface is where the electron probability density (and wave function) equals zero, meaning $\psi_{2s}(x) = 0$.

From the wave function graph in Figure 2, point B represents where the wave function crosses zero on the x-axis, indicating the location of the radial node.

Points A and C show maximum and minimum values respectively, while point D shows a non-zero region beyond the node.

14. (54) The energy of a spectral line in the hydrogen spectrum is given by $E = 13.6\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$ eV.

For the Lyman series, the lowest energy line L_1 corresponds to the transition from $n = 2$ to $n = 1$.

$E(L_1) = 13.6\left(\frac{1}{1^2} - \frac{1}{2^2}\right) = 13.6\left(1 - \frac{1}{4}\right) = 13.6 \times \frac{3}{4}$ eV.

For the Balmer series, the lowest energy line B_1 corresponds to the transition from $n = 3$ to $n = 2$.

$E(B_1) = 13.6\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = 13.6\left(\frac{1}{4} - \frac{1}{9}\right) = 13.6 \times \frac{5}{36}$ eV.

Given $E(L_1) = x \cdot E(B_1)$, we have:

$13.6 \times \frac{3}{4} = x \times 13.6 \times \frac{5}{36}$

$\frac{3}{4} = x \times \frac{5}{36}$

$x = \frac{3}{4} \times \frac{36}{5} = 3 \times \frac{9}{5} = \frac{27}{5} = 5.4$

The value of x is $5.4 = 54 \times 10^{-1}$.

Comparing with the required format $x = \text{integer} \times 10^{-1}$, the integer is 54.